Arithmetic Quantum Cat Maps:
Exact Fourth-Moment Reductions, Cyclotomic Row Support, and Heisenberg-Scale Trace Criteria

Maksim Churilov

Research draft — 5 September 2026

Abstract

We develop exact arithmetic reductions for a fourth-moment problem in quantum cat maps and isolate two logically separate closure problems before microscopic Bohigas–Giannoni–Schmit (BGS) universality can be addressed. Let p be an odd prime, let $H \le \mathbf{F}_p^\times$ have odd prime order K, and put

$$c(h) = \frac{h+1}{h-1}, \qquad A = c(H \setminus \{1\}).$$

An exact pair-key reduction gives

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2 \mathbf{1}_{t=0},$$

and both nontrivial fluctuation layers are controlled by the additive energy $E_+(A)$. A source-exact subgroup-curve theorem yields

$$E_+(A) < 34K^{8/3} \qquad (100 < K < p^{3/4}/3),$$

while the corrected off/off elliptic analysis gives

$$\mathrm{Var}_{t\neq 0}\, O(t) \ll pE_+(A) + K^2 p + K^4 p^{1/2} + K^4.$$

The pointwise arithmetic obstruction admits a sharper exact geometry. A fixed-product decomposition reduces non-forced excess to intersections of special affine copies of the prime cyclotomic trace set. Fixed-degree subresultants in the maximal real cyclotomic field define an exact split-prime order-three support ${\Lambda'_K}^{(3)}$, and outside that support

$$E_+(A) \le 11(K-1)(K-2).$$

For the inverse row ratio we prove a reflection identity that halves the polynomial degree. More generally, Bass's prime-conductor relations for cyclotomic units imply a complete characteristic-zero affine-intercept classification: every intercept fibre has cardinality at most two, and every double fibre is explicit. Thus order-three intersections are genuinely created by finite reduction.

We then replace the many-coefficient subresultant support by a scalar shift-resultant

$$\mathcal{R}_{K,r}(Z) = \mathrm{Res}_T(P_1(T), R_r(\alpha_r T + \beta_r + Z)) = c_0 + c_1 Z + c_2 Z^2 + \cdots .$$

For every split embedding prime, the order of vanishing at $Z = 0$ is exactly the row-intersection number. Consequently $I \ge 3$ is equivalent to $c_0 = c_1 = c_2 = 0$ modulo one prime ideal. Individual intersections are also congruences between explicit products of cyclotomic units. For each fixed row ratio the supported split primes satisfy the deterministic logarithmic budget

$$\sum_{p\equiv 1\,(K),\ I_r(p)\ge 3} \log p = O(K^3).$$

This does not yet prove the required polynomial largest-prime bound: summing over $O(K)$ ratios returns the existing $O(K^4)$ global scale.

A companion minimal-extension theorem uses Frobenius: if $f = \mathrm{ord}_K(p)$ and the order-K subgroup is placed in $\mathbf{F}_{p^f}$, every non-forced collision orbit has size exactly f. Together with the global resultant budget this gives a density-one exact baseline for unrestricted prime pairs $K \asymp X^\theta$, $p \asymp X$ for every fixed $\theta < 1/3$; it does not improve the split slice, where $f = 1$.

Finally, a partition-Möbius identity and Poisson summation give an exact trace-to-local-statistics bridge: microscopic band-limited k-point statistics require joint traces at powers of order the matrix dimension, the Heisenberg scale. The remaining spectral theorem is therefore a weighted Heisenberg-scale trace comparison for the correct desymmetrized block, plus a high-frequency tail estimate beyond band-limited tests. No microscopic BGS/GOE/COE universality theorem is claimed.

1 Scope and evidence discipline

Throughout the split finite-field part, p is an odd prime and $H \le \mathbf{F}_p^\times$ is a multiplicative subgroup of odd prime order K. Division is in $\mathbf{F}_p$. Exact statements are promoted only with their hypotheses and load-bearing derivations. Finite computations are certificates, regressions or falsifiers, never substitutes for growing-parameter proofs. When a theorem from the literature is used, the relevant source and hypothesis are identified.

Moment information and microscopic spectral statistics are kept separate. Arithmetic cat-map literature contains strong quantum-ergodicity and matrix-element results [1, 2, 3, 4, 5], while random-matrix universality is rigorous in special model classes such as mixing quantum graphs under their own assumptions [6, 7]. None of those results is silently transferred to the arithmetic family studied here.

2 Exact pair-key geometry and fourth-moment coupling

For $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every such (u, v) gives a unique admissible off-diagonal pair. The diagonal $x = y$ collapses to $(u, v, d) = (1, 1, 0)$ with multiplicity $p - 2$.

Proof. The equations $x = uy$ and $x - 1 = v(y - 1)$ give $(u - v)y = 1 - v$, hence (2). The diagonal statement is immediate. □

Corollary 2.2. *Under the subgroup constraints $u, v \in H$, the number of admissible off-diagonal ordered pairs is exactly $(K - 1)(K - 2)$.*

Give each collapsed key $r = (u_r, v_r, d_r)$ multiplicity n_r, equal to one off the diagonal and $p - 2$ at $(1, 1, 0)$. Define

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \mathbf{1}_H(v_r v_s) \mathbf{1}_{d_r + d_s = t}. \tag{3}$$

Let $O(t)$ be the off/off contribution and

$$C_H(t) = \#\left\{(u,v) \in (H \setminus \{1\})^2 : u \neq v,\ \frac{(u-1)(v-1)}{v-u} = t\right\}.$$

Then

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2\mathbf{1}_{t=0}. \tag{4}$$

The coefficient two is load-bearing: diagonal/off and off/diagonal are distinct ordered classes.

Proposition 2.3 (Exact total coupled count)**.**

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K.$$

If

$$\Sigma_4(q) = \sum_{t \in \mathbf{F}_p} B(t)e_p(qt), \qquad V = \sum_t B(t), \qquad Z = B(0),$$

finite Fourier orthogonality gives

$$\mathbb{E}_{q\neq 0}\Sigma_4(q) = \frac{pZ - V}{p-1}, \qquad \mathrm{Var}_{q\neq 0}\,\Sigma_4(q) = p\,\mathrm{Var}_{t\neq 0}\,B(t). \tag{5}$$

No probabilistic independence assumption enters this reduction.

3 Cayley transform and the pointwise energy target

Put

$$c(h) = \frac{h+1}{h-1}, \qquad A = c(H \setminus \{1\}).$$

For $u \neq v$,

$$\frac{(u-1)(v-1)}{v-u} = \frac{2}{c(u)-c(v)}, \qquad c(h^{-1}) = -c(h). \tag{6}$$

Thus $A = -A$, $0 \notin A$, and the nonzero multiplicities of C_H are the nonzero additive-difference multiplicities of A up to the permutation $x \mapsto 2/x$. Consequently

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{7}$$

With $n = K-1$, identity and central inversion force the nonzero baseline

$$n(2n-3) = (K-1)(2K-5). \tag{8}$$

For the unnormalized Fourier transform,

$$E_+(A) = \frac{|A|^4}{p} + \frac{1}{p}\sum_{q\neq 0} |\widehat{1_A}(q)|^4. \tag{9}$$

Hence the sharp pointwise target is

$$E_+(A) = \frac{(K-1)^4}{p} + O(K^{2+o(1)}), \tag{10}$$

equivalently

$$\sum_{q\neq 0} |\widehat{1_A}(q)|^4 \ll pK^{2+o(1)}.$$

There are two useful exact reformulations. With $y = (h-1)^{-1}$,

$$h = \frac{y+1}{y}, \qquad c(h) = 1 + 2y,$$

so the problem is an additive-energy question for the reciprocal shifted subgroup $(H-1)^{-1}$. Solving $c(h) - c(k) = t$ gives

$$k = \frac{(2-t)h + t}{2 + t - th}, \tag{11}$$

which is the Cayley conjugate of a translation. Known shifted-subgroup and Möbius-incidence estimates [15, 16] do not by themselves imply (10).

4 A uniform pointwise energy bound

For $t \in \mathbf{F}_p$, write $r(t) = \#\{(a,b) \in A^2 : a + b = t\}$. If $a = c(x)$ and $b = c(y)$, clearing denominators gives

$$P_t(x,y) = (2-t)xy + t(x+y) - (2+t) = 0. \tag{12}$$

The bilinear determinant equals -4, so every fibre is absolutely irreducible in odd characteristic.

Theorem 4.1 (Direct Cayley energy bound)**.** *If*

$$100 < K < \frac{1}{3}p^{3/4},$$

then

$$E_+(A) < 34K^{8/3}. \tag{13}$$

Proof. For $t \neq 0, -2$, Theorem 2 of the source version of Vyugin–Makarychev [13], with its degree hypotheses checked on (12), gives $r(t) \leq 32K^{2/3}$. The exceptional fibres satisfy $r(0) = K - 1$ and $r(-2) \leq K - 1$. Since $\sum_t r(t) = (K-1)^2$,

$$E_+(A) \leq 2(K-1)^2 + 32K^{2/3} \sum_{t \neq 0,-2} r(t) < 34K^{8/3}.$$

□

The later journal version is [14]. Combining with (7) gives the mixed centered-variance bound

$$\mathrm{Var}_{t \neq 0}(2(p-2)C_H(t)) < 136pK^{8/3}. \tag{14}$$

5 Prime-order additive rigidity and square resultants

Let K be an odd prime and $\zeta = e^{2\pi i/K}$. Then

$$c(\zeta^a) = -i \cot \frac{\pi a}{K}. \tag{15}$$

Chowla's prime-modulus cotangent independence [8, 9] identifies the characteristic-zero Cayley set with a rational cross-polytope.

Theorem 5.1 (Prime-order additive rigidity)**.** *Every characteristic-zero equal-difference collision among* $\{c(\zeta^a) : a \in \mathbf{F}_K^*\}$ *is identity or central inversion-swap.*

For an admissible collision quadruple, clearing the Cayley denominators gives an integral polynomial

$$P(X) = (X-1)^3 Q(X), \qquad Q(X^{-1}) = X^{3-T} Q(X), \qquad \|P\|_1 \leq 12.$$

Theorem 5.2 (Square-resultant theorem)**.** *For every odd prime K and every admissible collision quadruple,*

$$\operatorname{Res}(\Phi_K, P) = K^3 M^2, \qquad M \in \mathbf{Z}. \tag{16}$$

For a non-forced characteristic-zero collision, $M \neq 0$.

Proof. Let $L = \mathbf{Q}(\zeta)$ and L^+ be its maximal real subfield. Choose m with $2m \equiv 3 - T \pmod{K}$. The reciprocal identity makes $\beta = \zeta^m Q(\zeta)$ a real algebraic integer. Hence

$$\mathrm{N}_{L/\mathbf{Q}}(Q(\zeta)) = \mathrm{N}_{L^+/\mathbf{Q}}(\beta)^2 = M^2.$$

Since $\mathrm{N}_{L/\mathbf{Q}}(\zeta - 1) = K$, taking norms of $P(\zeta) = (\zeta-1)^3 Q(\zeta)$ proves the formula. Non-vanishing follows from Theorem 5.1. □

The Fibonacci slice

$$\operatorname{Res}(\Phi_K(X), X^2 + 3X + 1) = F_K^2$$

shows that individual collision resultants can be exponentially large. At $K = 131$, $p = F_{131}$, independent arbitrary-precision computations together with a Lucas primality certificate give exact nonzero Cayley excess 32; this is a finite certificate, not an asymptotic theorem. The primitive-divisor input for the corresponding rational model is taken from [12].

6 Canonical difference discriminant and ratio orbits

Let

$$Q_K(X) = \frac{(X+1)^K - (X-1)^K}{2}, \qquad R_K(T) = \operatorname{Res}_X(Q_K(X), Q_K(X - T)).$$

Remove the diagonal factor T^{K-1} and take the primitive squarefree part $S_K(T)$. Then $\deg S_K = (K-1)^2/2$.

Theorem 6.1 (Single-discriminant classification)**.** *For every split prime $p \equiv 1 \pmod{K}$,*

$$\Delta_K(p) > 0 \iff p \mid \operatorname{Disc}(S_K), \qquad \Delta_K(p) \leq 4 v_p(\operatorname{Disc}(S_K)), \tag{17}$$

where

$$\Delta_K(p) = \sum_{d \neq 0} r_{A-A}(d)^2 - (K-1)(2K-5).$$

The polynomial factors into ratio orbits. If $[r]$ denotes the involution $r \sim r^{-1}$, then

$$S_K(T) = \prod_{[r]} F_r(T),$$

and Dedekind-sum separation of the orbit factors [17] yields exact squares in the cross-resultants.

Theorem 6.2 (Fourth-power discriminant structure)**.** *For a positive integer J_K,*

$$|\operatorname{Disc}(S_K)| = 2^{(K-1)^4/4} K^{(K-1)(K^2-7)/4} J_K^4. \tag{18}$$

Exact computations through the maintained range reject an earlier $O(K^3 \log K)$ residual-height heuristic: the cross-resultants retain genuine K^4-scale total archimedean mass. Thus (18) is structural, not a hidden proof of (10).

7 Fixed-product rows and exact real-trace support

For a valid subgroup pair put $u = xy$, $s = x + y$. Then

$$c(x) + c(y) = \frac{2(u-1)}{u-s+1}. \tag{19}$$

At fixed $u \neq 1$ and fixed nonzero Cayley parameter, there is at most one unordered pair. If $I_K(p)$ is the maximum number of common parameters of two distinct product rows, the exact cross-row decomposition gives

$$\Delta_K(p) \leq 4I_K(p)(K-1)(K-2). \tag{20}$$

Normalize a distinct row pair to $(1, r)$ and put

$$\tau = \zeta + \zeta^{-1}, \qquad \alpha_r = U_{r-1}(\tau/2), \qquad \beta_r = -2U_{r-2}(\tau/2).$$

The affine trace equation is

$$Y = \alpha_r X + \beta_r. \tag{21}$$

Let $C_K(T)$ be the minimal polynomial of τ and set

$$H_K(T) = (T-2)C_K(T).$$

The unique characteristic-zero common full trace is the invalid row trace $T = \tau$. Remove it:

$$P_{K,r}(T) = \frac{H_K(T)}{T-\tau}, \qquad Q_{K,r}(T) = \frac{H_K(\alpha_r T + \beta_r)}{T-\tau}. \tag{22}$$

Both are degree $m = (K-1)/2$ over $\mathbf{Z}[\tau]$. For each split embedding prime $\mathfrak{q} \mid p$,

$$\deg \gcd(P_{K,r}, Q_{K,r} \bmod \mathfrak{q}) = I_{1,r}(\mathfrak{q}). \tag{23}$$

Let $J^{\mathrm{tr}}_{K,r}$ be the coefficient ideal of the first three fixed-degree subresultants and write

$$J^{\mathrm{tr}}_{K,r} \cap \mathbf{Z} = \Lambda^{(3)}_{K,r}\mathbf{Z}.$$

Theorem 7.1 (Exact real-trace order-three support)**.** *For every split prime $p \equiv 1 \pmod{K}$,*

$$p \mid \Lambda^{(3)}_{K,r}$$

if and only if some row pair in ratio class $[r]$ has at least three common Cayley parameters.

Proof sketch. Equation (23) is exact because the valid trace cofactors remain fixed-degree and squarefree after split reduction. Brown–Traub's fixed-degree subresultant criterion [18] identifies gcd degree at least three with vanishing of the first three subresultants. Requiring all their coefficients to vanish in one prime ideal and contracting to $\mathbf{Z}$ gives the rational-prime statement. □

After removing factors $2, K$ and multiplying over ratio classes, write the support as ${\Lambda'_K}^{(3)}$. Then

$$p \nmid {\Lambda'_K}^{(3)} \Longrightarrow I_K(p) \leq 2, \quad \Delta_K(p) \leq 8(K-1)(K-2), \quad E_+(A) \leq 11(K-1)(K-2). \tag{24}$$

Earlier dehomogenized projective-row invariants D' and G' remain valid only as one-way obstructions: specialization can lower their leading degree at the projective point at infinity. The real-trace construction is the active exact geometric support.

8 Inverse-row reflection and half-degree support

The ratio $r = -1$ has an additional involution. Here

$$\alpha_{-1} = -1, \qquad \beta_{-1} = 2\tau,$$

and

$$Q(T) = -P(2\tau - T). \tag{25}$$

Write

$$F(Z) = P(\tau + Z) = E(Z^2) + ZO(Z^2).$$

Theorem 8.1 (Inverse-row halving)**.** *For every split embedding prime* $\mathfrak{q}$*,*

$$I_{-1}(\mathfrak{q}) = 2 \deg \gcd(E, O \bmod \mathfrak{q}). \tag{26}$$

Thus inverse-row intersections are even and the order-three event is a corank-two problem at roughly half the degree. A fixed-degree subresultant ideal defines an exact split support integer Ξ_K. The trace set is also the squarefree support of the Chebyshev fibre

$$2(T_K(T/2) - 1) = (T - 2)C_K(T)^2. \tag{27}$$

At $(K, p) = (31, 1303)$ a direct finite-field check gives $I_{-1}(p) = 4$.

9 Cyclotomic trace-intercept rigidity

We now classify the characteristic-zero fibres underlying all ratios. Put

$$t_a = \zeta^a + \zeta^{-a}, \qquad s_a = \zeta^a - \zeta^{-a},$$

and let $[a]$ denote the trace class $a \sim -a$. Define

$$\gamma_r(a, b) = t_b - \alpha_r t_a. \tag{28}$$

For $1 \le j \le m$ set

$$E_j = |1 - \zeta^j|.$$

Lemma 9.1 (Prime-order cyclotomic absolute values)**.** *The positive real numbers* $E_1, \dots, E_m$ *are multiplicatively independent.*

Proof. Bass's prime-power theorem for the group generated by $1-\zeta^j$, modulo torsion, identifies the prime-conductor Galois orbit modulo complex conjugation with a free $\mathbf{Z}$-basis [10]. If $\prod_j E_j^{n_j} = 1$, squaring and using

$$1 - \zeta^{-j} = -\zeta^{-j}(1 - \zeta^j)$$

produces a relation modulo torsion with exponent $2n_j$ on each basis element. Hence every $n_j = 0$. □

We use the elementary factorizations

$$|s_n| = E_{[2n]}, \qquad |t_x - t_y| = E_{[x-y]}E_{[x+y]} \tag{29}$$

for distinct trace classes.

Theorem 9.2 (Trace-intercept fibre rigidity)**.** *Let $K \geq 7$ be an odd prime and $r \neq 1$. Every fibre of*

$$([a],[b]) \longmapsto \gamma_r(a,b)$$

has cardinality at most two.

If $r = -1$, every double fibre is a swap. If $r \neq \pm 1$, every double fibre is, up to signs of representatives and exchange of the two points,

$$([a],[a+1-r]), \qquad ([a+2],[a+1+r]). \tag{30}$$

Moreover the equality in (30) is exact.

Proof. For $r = -1$, $\gamma_{-1}(a,b) = t_a + t_b$. Equality of two such sums is a rational relation among prime-order roots of unity with total coefficient sum zero. Since the only rational relation among $1, \zeta, \ldots, \zeta^{K-1}$ is a multiple of Φ_K, all coefficients must vanish; only identity or swap remains.

Assume $r \neq \pm 1$ and two distinct pairs have the same intercept. Then both source and target trace classes differ and

$$t_b - t_d = \alpha_r(t_a - t_c).$$

Taking absolute values and using (29) gives

$$E_{[b-d]}E_{[b+d]}E_{[2]} = E_{[a-c]}E_{[a+c]}E_{[2r]}.$$

Lemma 9.1 therefore yields the multiset identity

$$\{[b-d],[b+d],[2]\} = \{[a-c],[a+c],[2r]\}. \tag{31}$$

Since $[2r] \neq [2]$, the class $[2]$ must match one of $[a-c]$ or $[a+c]$. After changing representatives and exchanging the source points, take $c = a+2$. Cancelling $[2]$ in (31) gives

$$\{[b-d],[b+d]\} = \{[2r],[2a+2]\},$$

hence the target classes are exactly $[a+1-r]$ and $[a+1+r]$. The original signed equality selects the ordering. Finally

$$t_{x+r} - t_{x-r} = s_r s_x, \qquad t_{a+2} - t_a = s_1 s_{a+1}$$

proves the displayed double-fibre identity directly.

A fibre of size three would give three distinct source trace classes pairwise adjacent by a step of two. Multiplying indices by 2^{-1} identifies this graph with the quotient of the odd K-cycle by $a \sim -a$, which is a path (up to a harmless terminal loop) and has no triangle. Thus no third point can occur. □

Corollary 9.3. *The distinguished characteristic-zero row solution $([1],[r])$ is isolated. Thus every finite order-three intersection is genuinely created by reduction.*

The structure is related to the cyclotomic-unit mechanism used in projective equivalence of roots of unity [11], but the present theorem is a special prime-order affine statement proved directly above.

10 Cyclotomic-unit matching and scalar shift-resultant support

The same row equation can be rewritten as a congruence between explicit cyclotomic units. Let division by two in subscripts mean multiplication by 2^{-1} modulo K and put

$$u_n = \frac{s_n}{s_1}.$$

For valid classes $[a] \neq [1]$, $[b] \neq [r]$ define

$$W_r(b) = u_{(b+r)/2}u_{(b-r)/2}, \qquad V_r(a) = u_r u_{(a+1)/2}u_{(a-1)/2}. \tag{32}$$

The difference identities give

$$\gamma_r(a,b) - \beta_r = s_1^2(W_r(b) - V_r(a)). \tag{33}$$

Since s_1 is a unit away from K, for every split embedding prime $\mathfrak{q}$,

$$\gamma_r(a,b) \equiv \beta_r \pmod{\mathfrak{q}} \iff W_r(b) \equiv V_r(a) \pmod{\mathfrak{q}}. \tag{34}$$

There is a second exact support detector involving only three scalar algebraic integers. Let

$$P_1(T) = \frac{H_K(T)}{T - t_1}, \qquad R_r(U) = \frac{H_K(U)}{U - t_r}$$

and define

$$\mathcal{R}_{K,r}(Z) = \mathrm{Res}_T(P_1(T), R_r(\alpha_r T + \beta_r + Z)) = c_0 + c_1 Z + c_2 Z^2 + \cdots. \tag{35}$$

Theorem 10.1 (Scalar shift-resultant support)**.** *For every split embedding prime* $\mathfrak{q}$*,*

$$\mathrm{ord}_{Z=0}(\mathcal{R}_{K,r}(Z) \bmod \mathfrak{q}) = I_{1,r}(\mathfrak{q}). \tag{36}$$

Consequently

$$I_{1,r}(\mathfrak{q}) \geq 3 \iff c_0 \equiv c_1 \equiv c_2 \equiv 0 \pmod{\mathfrak{q}}. \tag{37}$$

Proof. Over the split residue field,

$$P_1(T) = \prod_{[a]\neq[1]} (T - t_a), \qquad R_r(U) = \prod_{[b]\neq[r]} (U - t_b).$$

The product formula for the resultant therefore gives

$$\mathcal{R}_{K,r}(Z) = \prod_{[a]\neq[1]} \prod_{[b]\neq[r]} (Z + \alpha_r t_a + \beta_r - t_b). \tag{38}$$

Every factor is linear in Z and vanishes at zero exactly for one valid row intersection, proving (36) and (37). □

Define the three-generator ideal

$$J_{K,r}^{\mathrm{shift}} = (c_0, c_1, c_2) \subset \mathbf{Z}[\tau], \qquad J_{K,r}^{\mathrm{shift}} \cap \mathbf{Z} = \Omega_{K,r}^{(3)}\mathbf{Z}.$$

Corollary 10.2 (Three-generator exact support)**.** *For split rational primes,* $\Omega_{K,r}^{(3)}$ *has exactly the same order-three prime support as* $\Lambda_{K,r}^{(3)}$*.*

Equality of the full integers or of non-split valuations is not asserted. Exact SymPy/Smith anchors give prime-to-$2K$ support $1, 1, 1, 5^2$ for $(K, r) = (7, 2), (11, 7), (13, 3), (13, 12)$, matching the corresponding real-trace support.

Theorem 10.3 (Per-ratio support mass)**.** *Uniformly in $r \neq 1$,*

$$\log \left| \mathrm{N}_{\mathbf{Q}(\tau)/\mathbf{Q}}(c_0) \right| = O(K^3), \tag{39}$$

and hence

$$\sum_{\substack{p \equiv 1 \,(K) \\ I_r(p) \geq 3}} \log p = O(K^3). \tag{40}$$

Proof. Corollary 9.3 gives $c_0 \neq 0$ in characteristic zero. Under the embedding $\zeta \mapsto \zeta^j$,

$$|u_n^{(j)}| = \frac{|\sin(2\pi jn/K)|}{|\sin(2\pi j/K)|}.$$

With $d_j = \min(j, K/2 - j)$, the elementary sine lower bound gives $|u_n^{(j)}| \leq K/(4d_j)$. Every factor of (38) at $Z = 0$ is therefore at most $2 + K/d_j$ in absolute value. Since there are m^2 factors,

$$\log |\,\mathrm{N}(c_0)| \leq m^2 \sum_{j=1}^{m} \log(2 + K/d_j).$$

The last sum is $O(K)$ by splitting at $K/4$ and applying the elementary Stirling estimate to the resulting integer and half-integer products. This proves (39). Every supported split prime divides the rational integer $|\,\mathrm{N}(c_0)|$, yielding (40). □

Summing over $O(K)$ ratio classes returns $O(K^4)$, so Theorem 10.3 does not prove a polynomial bound for the largest supported split prime. The strongest current pointwise closure target remains, for example,

$$P^+_{\mathrm{split}}({\Lambda'_K}^{(3)}) \leq K^{4+o(1)}, \tag{41}$$

or an equivalent theorem for the simultaneous scalar support. The new formulation shows that the high-value problem is common-factor control for c_0, c_1, c_2 or the explicit unit congruences (34), not another height estimate on c_0 alone.

11 Global split-prime collision budget

Simultaneous multiplication of all four exponent indices by $a \in \mathbf{F}_K^\times$ acts freely on non-forced collision quadruples. The number of normalized scaling orbits is

$$N_K = K^3 - 5K^2 + 6K + 1. \tag{42}$$

For one orbit let P_q be its collision polynomial and $R_q = \mathrm{Res}(\Phi_K, P_q)$.

Theorem 11.1 (Weighted split-prime budget)**.** *For every odd prime K,*

$$\sum_{p \equiv 1 \,(K)} \Delta_K(p) \log p \leq N_K\big((K-1)\log 12 - 3\log K\big). \tag{43}$$

Proof. For one split prime, the number of scaled members of an orbit that collide is bounded by $v_p(R_q)$. Since $R_q = K^3 M_q^2$ and $|R_q| \leq 12^{K-1}$, summing the valuations over normalized orbits gives the claim. □

12 Frobenius amplification over minimal finite extensions

Let $p \neq 2, K$ be any rational prime and put

$$f = \mathrm{ord}_K(p), \qquad q = p^f.$$

Then $\mathbf{F}_q^\times$ contains the unique subgroup H of order K; put $A_{K,p} = c(H \setminus \{1\})$ and let $\Delta_K^{\mathrm{ext}}(p)$ denote the non-forced nonzero equal-difference excess.

Theorem 12.1 (Frobenius amplification)**.** *Every non-forced collision orbit under the p-power Frobenius has size exactly f. In particular,*

$$f \mid \Delta_K^{\mathrm{ext}}(p). \tag{44}$$

Proof. Frobenius sends the exponent tuple (i, j, r, s) to (pi, pj, pr, ps) modulo K. If a positive power $d < f$ fixed the tuple, any nonzero coordinate would give $p^d \equiv 1 \pmod K$, contradicting the definition of f. □

The normalized resultant argument works at one prime ideal over p, so the split budget and discriminant classification extend to all $p \neq 2, K$:

$$\sum_{p \neq 2, K} \Delta_K^{\mathrm{ext}}(p) \log p \leq N_K\big((K-1)\log 12 - 3\log K\big), \tag{45}$$

$$\Delta_K^{\mathrm{ext}}(p) > 0 \iff p \mid \mathrm{Disc}(S_K), \qquad \Delta_K^{\mathrm{ext}}(p) \leq 4 v_p(\mathrm{Disc}(S_K)). \tag{46}$$

Theorem 12.2 (Density-one minimal-extension baseline)**.** *Let $K \asymp X^\theta$ and $p \asymp X$ range independently over primes. For every fixed $0 < \theta < 1/3$, asymptotically almost every pair satisfies*

$$\Delta_K^{\mathrm{ext}}(p) = 0, \qquad E_+(A_{K,p}) = 3(K-1)(K-2)$$

exactly.

Proof sketch. Choose β with $\max(0, 4 - 1/\theta) < \beta < 1$. Residue classes modulo K of order at most K^β number at most $K^\beta \tau(K-1) = K^{\beta + o(1)}$; Brun–Titchmarsh and PNT make their pair density $O(K^{\beta - 1 + o(1)})$. On the complement, every bad prime costs at least K^β units of excess by Theorem 12.1; (45) then gives density $O(K^{4-\beta}/X)$. Both terms vanish by the choice of β. □

For the original split family, $p \equiv 1 \pmod K$ gives $f = 1$, so this amplification disappears.

13 Full Jacobi trace and averaged split theorems

Let $S = H^\perp$ and define the normalized annihilator Jacobi matrix

$$J_S(q)_{\chi,\psi} = \sum_{x \in \mathbf{F}_p \setminus \{0,1\}} \chi^{-1}(x)\psi(x-1)e_p(qx), \qquad M_S(q) = \frac{e_p(-q/2)}{p-1} J_S(q).$$

Theorem 13.1 (Cayley–Jacobi trace identity)**.** *For every $r \in \mathbf{F}_p$,*

$$\widehat{1_A}(r) = K \,\mathrm{Tr}\, M_S(2r). \tag{47}$$

Consequently

$$\sum_q |\,\mathrm{Tr}\, M_S(q)|^2 = \frac{p(K-1)}{K^2}, \tag{48}$$

$$\sum_q |\,\mathrm{Tr}\, M_S(q)|^4 = \frac{pE_+(A)}{K^4}. \tag{49}$$

For nonzero q, the exact moments are

$$\mu_2(K,p) = \frac{(K-1)(p-K+1)}{K^2(p-1)}, \tag{50}$$

$$\mu_4(K,p) = \frac{pE_+(A) - (K-1)^4}{K^4(p-1)}. \tag{51}$$

For the dyadic split family with prime $K \asymp X^\theta$ and $p \asymp X$, Bombieri–Vinogradov and PNT [19, 20] give:

- for fixed $\theta < 1/3$,
$$\text{average}\, \frac{\Delta_K(p)}{K^2} \ll_\theta X^{3\theta-1} \longrightarrow 0;$$
- for fixed $\theta < 1/5$, asymptotically almost every split pair has
$$\Delta_K(p) = 0, \qquad E_+(A) = 3(K-1)(K-2)$$
exactly.

The normalized full Jacobi-trace fourth/second moment ratio tends to 3 in the same $\theta < 1/3$ double average. This is not a CLT and is not a local spectral law.

14 Corrected off/off elliptic correlation

For an off-diagonal key (u, v) set

$$d(u,v) = \frac{(u-1)(v-1)}{v-u}.$$

For a generic parameter $\lambda = (h_1, h_2) \in H^2$, the corresponding one-parameter fibre is a non-isotrivial genus-one pencil. Its distinguished reducible fibre occurs at

$$t_\lambda = \frac{2}{c(h_1) - c(h_2)}$$

and contains exactly

$$2p - 8 \tag{52}$$

valid affine points. After extracting this coherent spike, Igusa geometric monodromy [22] and the Fouvry–Kowalski–Michel trace-correlation argument [21] give square-root quasi-orthogonality for distinct geometric parameter classes.

Theorem 14.1 (Corrected off/off variance)**.** *For $p > 3$ and odd $|H| = K$,*

$$\mathrm{Var}_{t\neq 0}\, O(t) \ll pE_+(A) + K^2p + K^4p^{1/2} + K^4, \tag{53}$$

with an absolute implied constant.

Combining with Theorem 4.1,

$$\mathrm{Var}_{t\neq 0}\, O(t) \ll pK^{8/3} + K^4p^{1/2} + K^2p + K^4.$$

Thus in $K = o(p^{1/4})$ the elliptic layer has collapsed onto the same $K^{2/3}$ pointwise Cayley-energy loss as the mixed layer.

15 Exact trace-to-local-statistics bridge

Let U be an $N \times N$ unitary matrix with eigenvalues $e^{2\pi i\theta_1}, \dots, e^{2\pi i\theta_N}$. For integers $m_1 + \cdots + m_k = 0$, define

$$D_U(m_1, \dots, m_k) = \sum_{j_1,\dots,j_k \text{ distinct}} e^{2\pi i \sum_{a=1}^{k} m_a \theta_{j_a}}.$$

For a partition π of $\{1, \dots, k\}$, put

$$\mu(\pi) = \prod_{B\in\pi} (-1)^{|B|-1}(|B|-1)!, \qquad T_U(n) = \operatorname{Tr}(U^n), \quad T_U(0) = N.$$

Theorem 15.1 (Factorial trace identity)**.**

$$D_U(m_1, \dots, m_k) = \sum_{\pi} \mu(\pi) \prod_{B\in\pi} T_U\left(\sum_{a\in B} m_a\right). \tag{54}$$

Proof. Impose equality of eigenvalue indices inside each block of a set partition and apply Möbius inversion on the partition lattice. □

For $\phi \in \mathcal{S}(\mathbf{R}^{k-1})$, periodize at the mean-spacing scale,

$$\Phi_N(t) = \sum_{n\in\mathbf{Z}^{k-1}} \phi(N(t+n)),$$

and define the translation-invariant factorial statistic

$$C_{k,N}(U;\phi) = \frac{1}{N} \sum_{j_1,\dots,j_k \text{ distinct}} \Phi_N(\theta_{j_1} - \theta_{j_k}, \dots, \theta_{j_{k-1}} - \theta_{j_k}).$$

Theorem 15.2 (Trace-to-local-statistics identity)**.** *With $m_k = -(m_1 + \cdots + m_{k-1})$,*

$$C_{k,N}(U;\phi) = N^{-k} \sum_{m\in\mathbf{Z}^{k-1}} \widehat{\phi}(m/N) D_U(m_1, \dots, m_k). \tag{55}$$

Proof. Apply Poisson summation to Φ_N and interchange the absolutely convergent Schwartz sum with the finite eigenvalue sum. □

For $k = 2$,

$$C_{2,N}(U;\phi) = N^{-2} \sum_{m\in\mathbf{Z}} \widehat{\phi}(m/N)(|\operatorname{Tr} U^m|^2 - N). \tag{56}$$

If $\widehat{\phi}$ is compactly supported, the relevant powers are $|m| = \Theta(N)$: microscopic correlations live at Heisenberg scale.

Corollary 15.3 (Weighted Heisenberg trace criterion)**.** *Let $\mathbb{E}_N$ be the averaging functional for the quantum system and $\mathbb{E}_N^{\mathrm{RMT}}$ the target random-matrix average. If for a fixed band-limited test*

$$N^{-k} \sum_{m} |\widehat{\phi}(m/N)| \left|\mathbb{E}_N D_U(m) - \mathbb{E}_N^{\mathrm{RMT}} D_V(m)\right| \longrightarrow 0,$$

then the corresponding averaged microscopic factorial statistic converges to its RMT value.

For arbitrary Schwartz tests one also needs a uniform high-frequency tail estimate. The deterministic Fourier bridge is closed; the hard spectral task is the Heisenberg-scale joint-trace comparison for the correct desymmetrized block.

16 Reproducibility and adversarial controls

The repository deliberately uses mechanically different verification routes.

- Python/SymPy regenerates exact resultants, discriminants, Smith contractions, finite-extension Frobenius checks, Jacobi identities and elliptic spike algebra.
- C++20 independently checks finite-field hot paths, arbitrary-precision witnesses and the sparse group-ring characteristic-zero intercept theorem.
- Lean 4 kernel-checks finite pair-key/Cayley statements, historical falsifiers and bounded exact certificates.
- CI rebuilds the maintained code and recompiles this article; a separate workflow synchronizes the committed PDF from the TeX source.

The C++ intercept regression checks every normalized ratio in the maintained quick range through $K = 199$; a recorded extended run also checks prime $K = 251, 307, 401, 503, 601, 701, 809$, always with maximum fibre size two and the predicted double-fibre form. This verifies the implementation, not Theorem 9.2, whose proof is independent.

Exact scalar shift-resultant/Smith anchors are

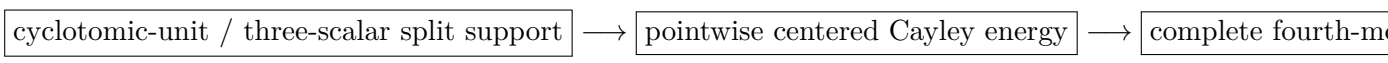
(K, r)	$\Omega^{(3)}_{K,r}$	prime-to-$2K$ part
$(7, 2)$	7^3	1
$(11, 7)$	11^5	1
$(13, 3)$	13^6	1
$(13, 12)$	$2^4\,5^2\,13^6$	5^2.

Their prime-to-$2K$ support agrees with the corresponding exact real-trace Λ' anchors; equality of full valuations is not claimed.

Adversarial finite controls include $(K, p) = (29, 728017)$ and $(31, 1042903)$ with $p > K^4$ but positive excess, disproving a bare "$p > K^4$ implies baseline" rule; $(K, p) = (41, 2908049)$ has a row pair with two common parameters, disproving $I_K(p) \le 1$; and $(K, p) = (31, 1303)$ has an inverse-row intersection of size four.

17 The remaining BGS frontier

The arithmetic fourth-moment chain is now

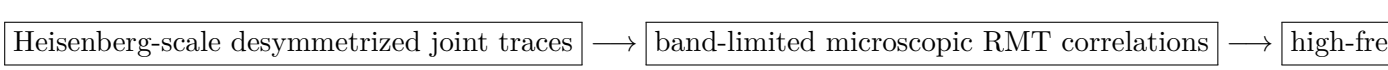


The exact order-three event has two specialization-safe encodings: the fixed-degree Λ subresultants and the three shift coefficients c_0, c_1, c_2. The new per-ratio $O(K^3)$ logarithmic mass is not a polynomial largest-prime bound. A sufficient remaining target is

$$P^+_{\mathrm{split}}({\Lambda'_K}^{(3)}) \le K^{4+o(1)},$$

or an equivalent simultaneous-gcd theorem for the scalar support.

The microscopic spectral chain is independently



The middle Fourier arrow is exact by Theorems 15.1–15.2; the load-bearing open problem is the trace estimate itself. Fixed powers, finitely many moments, or even a completed fourth-moment theorem cannot be promoted to microscopic BGS.

18 Conclusion

The present programme has reduced a complicated fourth-moment problem to a sharper arithmetic obstruction. The pair-key geometry is exact; the mixed and corrected off/off layers are governed by the same Cayley additive energy; prime-order characteristic-zero collisions are rigid; and exceptional split reductions admit canonical resultant/discriminant descriptions. Fixed-product rows give an exact order-three support in the maximal real cyclotomic field. The new characteristic-zero intercept theorem proves that each affine intercept fibre has at most two points and classifies all double fibres. The finite order-three event is therefore a genuine reduction phenomenon.

The same event is now encoded by only three scalar coefficients of one shift resultant, while each row intersection is a congruence between explicit cyclotomic units. This gives an $O(K^3)$ logarithmic support budget for each ratio but does not yet control the largest supported split prime polynomially. The next arithmetic target is simultaneous common-factor cancellation, not another generic incidence estimate.

Frobenius amplification gives a complementary density-one exact theorem over minimal finite extensions, while leaving the split slice unchanged. On the spectral side, the exact partition-Möbius/Poisson formula identifies Heisenberg-scale joint traces as the required input for microscopic correlations. Full BGS/GOE/COE universality remains open.

References

[1] P. Kurlberg and Z. Rudnick, *On the distribution of matrix elements for the quantum cat map*, Ann. of Math. **161** (2005), 489–507, doi:10.4007/annals.2005.161.489.

[2] P. Kurlberg, L. Rosenzweig and Z. Rudnick, *Matrix elements for the quantum cat map: fluctuations in short windows*, arXiv:math/0701685 (2007).

[3] S. Dyatlov and M. Jézéquel, *Semiclassical Measures for Higher-Dimensional Quantum Cat Maps*, Ann. Henri Poincaré **25** (2024), 1545–1605, doi:10.1007/s00023-023-01309-x.

[4] P. Kurlberg, A. Ostafe, Z. Rudnick and I. E. Shparlinski, *On Quantum Ergodicity for Higher Dimensional Cat Maps*, Commun. Math. Phys. **406** (2025), 174, doi:10.1007/s00220-025-05350-1.

[5] S. Bhakta and I. E. Shparlinski, *On quantum ergodicity for higher-dimensional cat maps modulo prime powers*, Bull. London Math. Soc. **58** (2026), doi:10.1112/blms.70300.

[6] Z. Pluhař and H. A. Weidenmüller, *Universal Quantum Graphs*, Phys. Rev. Lett. **112** (2014), 144102, doi:10.1103/PhysRevLett.112.144102.

[7] Z. Pluhař and H. A. Weidenmüller, *Quantum graphs and random-matrix theory*, J. Phys. A **48** (2015), 275102, doi:10.1088/1751-8113/48/27/275102.

[8] S. Chowla, *The nonexistence of nontrivial linear relations between the roots of a certain irreducible equation*, J. Number Theory **2** (1970), 120–123, doi:10.1016/0022-314X(70)90012-0.

[9] K. Wang, *On a theorem of S. Chowla*, J. Number Theory **15** (1982), 1–4, doi:10.1016/0022-314X(82)90078-6.

[10] H. Bass, *Generators and Relations for Cyclotomic Units*, Nagoya Math. J. **27** (1966), 401–407, doi:10.1017/S0027763000026258.

[11] H. Fu, *Projective Equivalence for the Roots of Unity*, J. Algebraic Combin. **56** (2022), 823–871, arXiv:1905.01510.

[12] M. Yabuta, *A Simple Proof of Carmichael's Theorem on Primitive Divisors*, Fibonacci Quart. **39** (2001), 439–443, doi:10.1080/00150517.2001.12428701.

[13] I. V. Vyugin and S. Makarychev, *Solutions of polynomial equation over* $\mathbf{F}_p$ *and new bounds of additive energy*, arXiv:1504.01354v2 (2015).

[14] S. Makarychev and I. Vyugin, *Solutions of Polynomial Equations in Subgroups of* $\mathbf{F}_p^*$, Arnold Math. J. **5** (2019), 105–121, doi:10.1007/s40598-019-00112-z.

[15] S. Macourt, I. D. Shkredov and I. E. Shparlinski, *Multiplicative Energy of Shifted Subgroups and Bounds On Exponential Sums with Trinomials in Finite Fields*, Canad. J. Math. **70** (2018), 1319–1338, doi:10.4153/CJM-2017-044-2.

[16] M. Rudnev and J. Wheeler, *On incidence bounds with Möbius hyperbolae in positive characteristic*, Finite Fields Appl. **78** (2022), 101978, doi:10.1016/j.ffa.2021.101978.

[17] S. Jabuka, S. Robins and X. Wang, *When are two Dedekind sums equal?*, Int. J. Number Theory **7** (2011), 2197–2203, arXiv:1103.0370.

[18] W. S. Brown and J. F. Traub, *On Euclid's Algorithm and the Theory of Subresultants*, J. ACM **18** (1971), 505–514, doi:10.1145/321662.321665.

[19] E. Bombieri, *On the large sieve*, Mathematika **12** (1965), 201–225, doi:10.1112/S0025579300005313.

[20] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications **53**, 2004, doi:10.1090/coll/053.

[21] É. Fouvry, E. Kowalski and P. Michel, *Counting sheaves using spherical codes*, Math. Res. Lett. **20** (2013), 305–323, doi:10.4310/MRL.2013.V20.N2.A8, arXiv:1210.0851.

[22] A. Bandini, I. Longhi and S. Vigni, *Torsion points on elliptic curves over function fields and a theorem of Igusa*, Expo. Math. **27** (2009), 175–209, doi:10.1016/j.exmath.2008.06.001, arXiv:0804.1425.